

5.24 Recall that changes in atoms and nuclei can	
a generate radiations over a wide frequency range	
b be caused by absorption of a range of radiations	

Use of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).
- Apply the relationships between frequency and wavelength across the electromagnetic spectrum (1a, 1c, 3c).
- Construct two-dimensional ray diagrams to illustrate reflection and refraction (qualitative – equations not needed) (5a, 5b).

Suggested practicals

- Investigate total internal reflection using a semi-circular block (glass or plastic).
- Construct devices using two converging lenses of differing focal lengths.
- Construct a simple spectrometer, from a CD or DVD, and use it to analyse common light sources.
- Investigate the areas beyond the visible spectrum, such as the work of Herschel and Ritter in discovering IR and UV respectively.

Topic 6 – Radioactivity

Students should:	Maths skills
6.1 Describe an atom as a positively charged nucleus, consisting of protons and neutrons, surrounded by negatively charged electrons, with the nuclear radius much smaller than that of the atom and with almost all of the mass in the nucleus	5b
6.2 Recall the typical size (order of magnitude) of atoms and small molecules	
6.3 Describe the structure of nuclei of isotopes using the terms atomic (proton) number and mass (nucleon) number and using symbols in the format using symbols in the format $^{13}_{6}\text{C}$	1a 3a
6.4 Recall that the nucleus of each element has a characteristic positive charge, but that isotopes of an element differ in mass by having different numbers of neutrons	2g 5b
6.5 Recall the relative masses and relative electric charges of protons, neutrons, electrons and positrons	
6.6 Recall that in an atom the number of protons equals the number of electrons and is therefore neutral	
6.7 Recall that in each atom its electrons orbit the nucleus at different set distances from the nucleus	5b
6.8 Explain that electrons change orbit when there is absorption or emission of electromagnetic radiation	5b
6.9 Explain how atoms may form positive ions by losing outer electrons	5b
6.10 Recall that alpha, β^- (beta minus), β^+ (positron), gamma rays and neutron radiation are emitted from unstable nuclei in a random process	
6.11 Recall that alpha, β^- (beta minus), β^+ (positron) and gamma rays are ionising radiations	
6.12 Explain what is meant by background radiation	
6.13 Describe the origins of background radiation from Earth and space	
6.14 Describe methods for measuring and detecting radioactivity limited to photographic film and a Geiger–Müller tube	
6.15 Recall that an alpha particle is equivalent to a helium nucleus, a beta particle is an electron emitted from the nucleus and a gamma ray is electromagnetic radiation	
6.16 Compare alpha, beta and gamma radiations in terms of their abilities to penetrate and ionise	

Students should:	Maths skills
6.17 Describe how and why the atomic model has changed over time including reference to the plum pudding model and Rutherford alpha particle scattering leading to the Bohr model	5b
6.18 Describe the process of β^- decay (a neutron becomes a proton plus an electron)	1b, 1c, 3c
6.19 Describe the process of β^+ decay (a proton becomes a neutron plus a positron)	1b, 1c, 3c
6.20 Explain the effects on the atomic (proton) number and mass (nucleon) number of radioactive decays (α , β , γ and neutron emission)	1b, 1c, 3c
6.21 Recall that nuclei that have undergone radioactive decay often undergo nuclear rearrangement with a loss of energy as gamma radiation	
6.22 Use given data to balance nuclear equations in terms of mass and charge	1b, 1c, 3c
6.23 Describe how the activity of a radioactive source decreases over a period of time	2g 4c
6.24 Recall that the unit of activity of a radioactive isotope is the Becquerel, Bq	
6.25 Explain that the half-life of a radioactive isotope is the time taken for half the undecayed nuclei to decay or the activity of a source to decay by half	1c, 1d 2a
6.26 Explain that it cannot be predicted when a particular nucleus will decay but half-life enables the activity of a very large number of nuclei to be predicted during the decay process	1c, 3d
6.27 Use the concept of half-life to carry out simple calculations on the decay of a radioactive isotope, including graphical representations	1a, 1b, 1c, 1d 2a, 2g 3a, 3b, 3c, 3d
6.28P Describe uses of radioactivity, including: a household fire (smoke) alarms b irradiating food c sterilisation of equipment d tracing and gauging thicknesses e diagnosis and treatment of cancer	
6.29 Describe the dangers of ionising radiation in terms of tissue damage and possible mutations and relate this to the precautions needed	
6.30P Explain how the dangers of ionising radiation depend on half-life and relate this to the precautions needed	

Students should:	Maths skills
6.31 Explain the precautions taken to ensure the safety of people exposed to radiation, including limiting the dose for patients and the risks to medical personnel	
6.32 Describe the differences between contamination and irradiation effects and compare the hazards associated with these two	
6.33P Compare and contrast the treatment of tumours using radiation applied internally or externally	5b
6.34P Explain some of the uses of radioactive substances in diagnosis of medical conditions, including PET scanners and tracers	
6.35P Explain why isotopes used in PET scanners have to be produced nearby	
6.36P Evaluate the advantages and disadvantages of nuclear power for generating electricity, including the lack of carbon dioxide emissions, risks, public perception, waste disposal and safety issues	
6.37P Recall that nuclear reactions, including fission, fusion and radioactive decay, can be a source of energy	
6.38P Explain how the fission of U-235 produces two daughter nuclei and the emission of two or more neutrons, accompanied by a release of energy	1b, 1c, 3c
6.39P Explain the principle of a controlled nuclear chain reaction	
6.40P Explain how the chain reaction is controlled in a nuclear reactor, including the action of moderators and control rods	5b
6.41P Describe how thermal (heat) energy from the chain reaction is used in the generation of electricity in a nuclear power station	
6.42P Recall that the products of nuclear fission are radioactive	
6.43P Describe nuclear fusion as the creation of larger nuclei resulting in a loss of mass from smaller nuclei, accompanied by a release of energy, and recognise fusion as the energy source for stars	1b, 1c, 3c
6.44P Explain the difference between nuclear fusion and nuclear fission	
6.45P Explain why nuclear fusion does not happen at low temperatures and pressures, due to electrostatic repulsion of protons	
6.46P Relate the conditions for fusion to the difficulty of making a practical and economic form of power station	

Uses of mathematics

- Make calculations using ratios and proportional reasoning to convert units and to compute rates (1c, 3c).
- Balance equations representing alpha-, beta- or gamma-radiations in terms of the masses and charges of the atoms involved (1b, 1c, 3c).
- **Calculate the net decline, expressed as a ratio, in a radioactive emission after a given number of half-lives (1c, 3d).**

Suggested practicals

- Investigate models which simulate radioactive decay.